

Opportunistic Infections in Adults

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Session Objectives

At the conclusion of the session/course, the student should be able to:

1. Discuss the various mechanisms of acquired immune deficiency.
2. Gauge the type and severity of immune deficiency in a patient presenting with opportunistic infection.
3. Differentiate opportunistic pathogens that may present in patients with immune deficiency.
4. Diagnose, treat and manage and patient with severe immune deficiency secondary to AIDS.
5. Identify the risk factors for immune reconstitution inflammatory syndrome.

Design

1. Definitions
2. Hematopoietic, Solid Organ Transplant and Myeloablation
3. HIV/AIDS
4. Opportunistic Organisms (Selected)

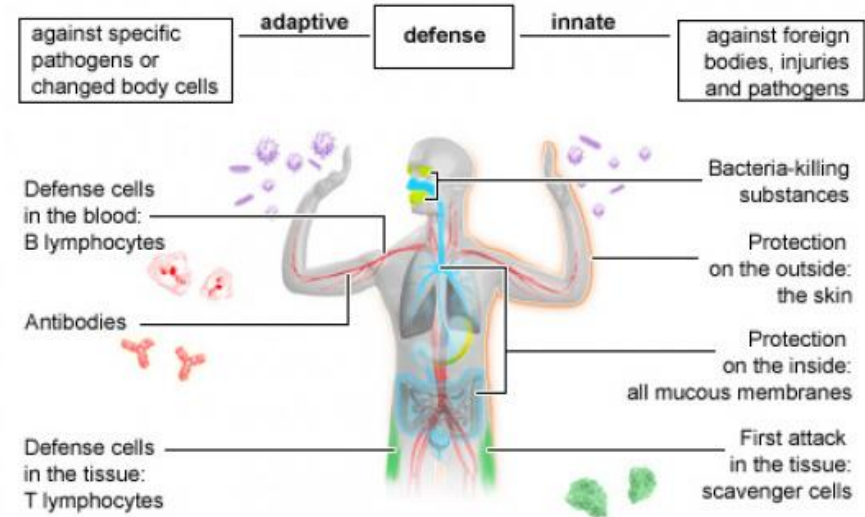
Each slide is a single concept

***Pause* points**

There is no way to completely cover this vast concept in a single lecture. You will have to research this subject on your own in order to fill the gaps.

What is immune deficiency?

- A defect of innate or adaptive immunity resulting in increased risk of:
 - Infection
 - Common, opportunistic pathogens and saprophytes
 - Newly acquired, commensal or reactivation
 - Malignancy
- Examples
 - Breaching of natural barriers
 - Advanced HIV/AIDS
 - Myelosuppressive chemotherapy
 - Immune suppression for
 - Autoimmune illness
 - Prevention of transplant rejection
 - Asplenia
 - Corticosteroids equivalent to prednisone ≥ 20 mg/day for ≥ 4 weeks



Question

Clostridium difficile colitis is a result of breaching what immune barrier?

1. Skin
2. Mucous membrane
3. Commensal flora
4. Humoral (antibody)

Question

Staphylococcus epidermidis central venous catheter-related bloodstream infection is the result of breaching what immune barrier?

1. Skin
2. Mucous membrane
3. Commensal flora
4. Humoral (antibody)

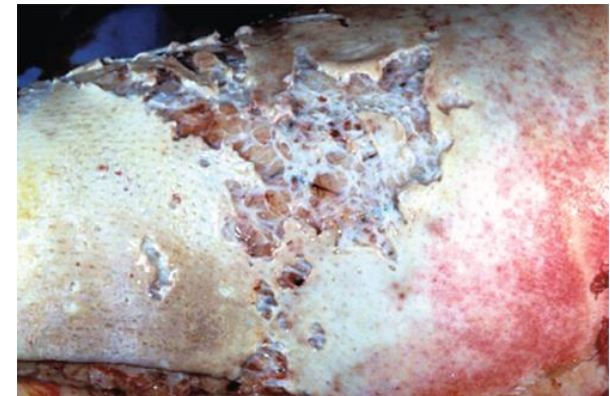
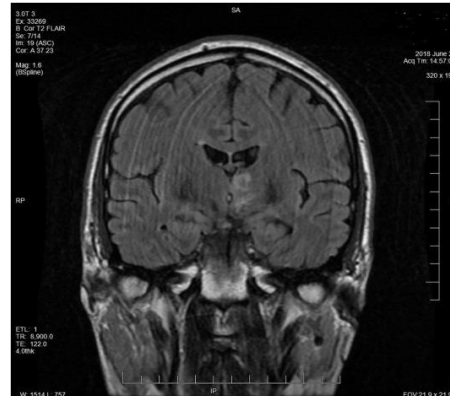
What is an Opportunistic Infection?

- An infection that does not typically occur in the immune competent host
 - Commensal organisms: *Staphylococcus epidermidis*
 - Microbiome imbalance: *Clostridium difficile*
 - Pharmaceuticals: *Lactobacillus*
 - Tap water: *Mycobacterium avium*, *Pseudomonas aeruginosa*, *Legionella pneumophila*
 - Reactivation: HSV, CMV, EBV, *Toxoplasma*, *M. tuberculosis*



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Question

Which of the following is a risk for recurrent *Neisseria* infections?

1. Systemic lupus erythematosus
2. Complement deficiency C5
3. Eculizumab
4. Idiopathic CD4 lymphopenia

Types of Immune Deficiencies

Deficiency	Examples	Organisms
Extremes of age	Birth, elderly	Zoster
Broad-spectrum abx	Sepsis Resistant organisms rx	Pseudomonas Candida
Foreign bodies	Central venous catheter Urinary catheter Joint replacement Heart valve replacement Pacemaker	<i>Staphylococcus epidermidis</i> , <i>S. aureus</i> , enterococci, <i>Corynebacterium</i> <i>Candida</i>
Trauma	Penetrating injury Surgery Burns	<i>S. epidermidis</i> , <i>S. aureus</i> , <i>Pseudomonas aeruginosa</i>
Chronic illness	Diabetes Chronic renal failure COPD, cystic fibrosis	<i>Staphylococcus aureus</i> , <i>S. epidermidis</i> , enteric Gram negatives, <i>Pseudomonas</i> , <i>Streptococcus pneumoniae</i> <i>Haemophilus influenzae</i> , NTMB
Complement	Properdin deficiency SLE Eculizumab	<i>Neisseria</i> infections

Question

Increased risk of infection with which of the following organisms occurs after autosplenectomy?

1. *Babesia*
2. *Streptococcus pneumoniae*
3. *Capnocytophaga*
4. *Staphylococcus epidermidis*

Types of Immune Deficiencies

Deficiency	Examples	Organisms
Alpha-4 (α 4) integrin	Natalizumab	PML
Humoral	Bruton's Multiple myeloma CLL Small lymphocyte lymphoma Splenectomy Common variable immunodeficiency	<i>Haemophilus influenzae</i> , <i>Neisseria meningitidis</i> , <i>Streptococcus pneumoniae</i> , <i>Capnocytophaga</i> Babesiosis
Granulocytopenia	Hematopoietic cell transplantation Myelosuppressive chemotherapy Acute leukemia	Gram-negative enterics, <i>Pseudomonas</i> , <i>Staphylococcus aureus</i> , <i>S. epidermidis</i> , Viridans streptococci <i>Candida</i> , <i>Aspergillus</i> , <i>Trichosporon</i> , <i>Scedosporium</i> , <i>Fusarium</i> , mucormycosis
Cellular immunity	HIV infection Lymphoreticular malignancies Corticosteroids, cyclosporine, tacrolimus Solid organ transplantation TNF inhibitors Idiopathic CD4 lymphopenia	<i>Listeria</i> , <i>Legionella</i> , <i>Salmonella</i> , <i>Mycobacterium</i> HSV, VZV, CMV, EBV, HBV, HCV <i>Cryptococcus</i> , <i>Coccidioides</i> , <i>Histoplasma</i> , <i>Pneumocystis</i> <i>Toxoplasma</i>

Question

Risk of infection by *Aspergillus* sp. Is increased in which phase of allogeneic hematopoietic stem cell transplant?

1. Pretransplantation
2. Pre-engraftment (1 – 21 days)
3. Post-engraftment (3 weeks – 3 months)
4. > 3 months

Hematopoietic Cell Transplantation

- Early post-transplant (pre-engraftment) 1-21 days
 - Severe neutropenia up to 21 days
 - Staphylococci, gram-negative, HSV, RSV, *Candida*, *Aspergillus*
- 3 weeks – 3 months (post-engraftment)
 - CMV, adenovirus, *Aspergillus*, *Candida*, *Pneumocystis*
- >3 months
 - VZV, *Aspergillus*, CMV

Question

Which of the following has the highest risk of CMV infection after solid organ transplant?

1. CMV positive donor, CMV positive recipient
2. CMV positive donor, CMV negative recipient
3. CMV negative donor, CMV negative recipient
4. CMV negative donor, CMV positive recipient

Solid Organ Transplantation

- Immediate postoperative
 - Infections involving transplanted organ
- 2-4 weeks posttransplant
 - Wound, IV catheter, UTI
 - Transplanted infections
- 1-6 months posttransplant
 - HSV, VZV, CMV, *Candida*, *Aspergillus*, *Cryptococcus*, *Pneumocystis*, *Listeria monocytogenes*, *Nocardia*, *Toxoplasma*

Risk of posttransplant CMV infection (1-6 months)

High risk—CMV Ab (–) recipient, CMV Ab (+) donor

Intermediate risk—CMV Ab (+) recipient,
CMV Ab (+) or Ab (–) donor

Lowest risk—CMV Ab (–) recipient, CMV Ab (–) donor

Question

Which of the following increases risk of infection in patients with absolute neutropenia?

1. Duration >7 days
2. Duration <7 days
3. Granulocyte count >500/ml
4. Granulocyte count >1000/ml

Myeloablative (cytotoxic) Chemotherapy

- +/- irradiation
- Mucosal barrier damage
 - Oral and gut organisms
- Neutropenia
 - Infection risk increases when neutrophils $<1000/\text{ml}$
 - “Absolute” neutropenia $<500/\text{ml}$
 - Extremely high-risk for infection $<100/\text{ml}$
 - Risk of infection increases with duration >7 days

Sources of Bacteria Commonly Infecting Neutropenic Patients

Skin	Oral Cavity	Gut
Coagulase-negative staphylococci	<i>Streptococcus viridans</i>	<i>Escherichia coli</i>
<i>Staphylococcus aureus</i>	Oral anaerobes	<i>Klebsiella</i> spp. <i>Enterobacter</i> spp. Other enteric bacteria Gut anaerobes Enterococci <i>Pseudomonas aeruginosa</i>

- Bobby, a 39-year-old male has fever, cough and SOB.

He was diagnosed with HIV/AIDS 10 years ago and was placed on ART. CD4 went from 150 to 775/ml and viral load was undetectable. He was lost to follow up 8 years ago.

He reported 3 weeks of progressively worse cough accompanied by at first fever less than 37.7 °C (100°F) for the last week as high as 39.4°C (103°F) accompanied by shaking chills. For the last 1½ weeks he has had progressive SOB, at first only on exertion and now at rest.

Bobby

He explains that 8 years ago he broke up with his partner of 4 years and returned to using injection drugs, mostly heroin. He stopped taking ART then and over the last year has been feeling unwell. He has lost 35 pounds and was hospitalized 8 months ago for 5 days with pneumonia. He has not otherwise seen a physician.

Exam is significant for:

A thin male in respiratory distress unable to finish full sentences.

RR 28, BP 93/58, HR 124, T 38.3°C (101°F), Wt 84 lbs., Ht 5'10"

He has no rash. White cottage cheese-like exudate is easily scraped off the buccal mucosa (consistent with oral candidiasis). Shoddy axillary nodes are present. Lung fields are clear, and percussion is normal. Thoracic paraspinous skin drag is prolonged at T1-T4 bilaterally.

He is admitted to the ICU, placed on mechanical ventilation. CBC shows pancytopenia. Chest x-ray shows bilateral interstitial and ground-glass opacities without effusions.

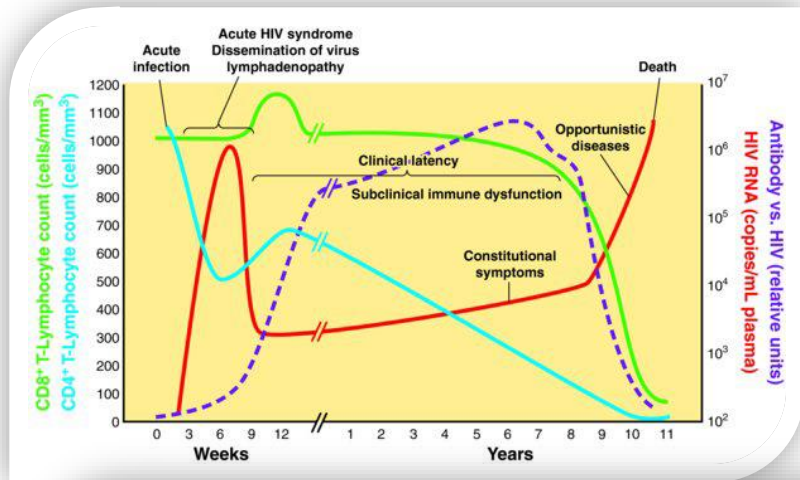
A procedure is done.

1. What procedure do you think was done?
2. What antibiotics would you start?

Pause

Advanced HIV/AIDS

• Humoral and cell-mediated immune deficiency



Stage	CD4 Count	CD4 %*	Clinical Evidence
Stage 0	Early HIV Infection		
Stage 1	≥500 cells/mm ³	≥26	No AIDS-defining condition
Stage 2	200-499 cells/mm ³	14-25	No AIDS-defining condition
Stage 3	<200 cells/mm ³	<14	or Documentation of AIDS-defining condition

CD4 Opportunistic Infections

<500 *Candida albicans* (thrush)
EBV (oral hairy leukoplakia)
Bartonella henselae
HHV-8
HPV

<200 *Pneumocystis jirovecii*
Histoplasma capsulatum
Progressive multifocal leukoencephalopathy (PML)

<100 Cerebral toxoplasmosis
Aspergillus fumigatus
Cryptosporidium
EBV-associated B-cell lymphoma

<50 CMV retinitis
Cryptococcal meningitis
Esophageal candidiasis
M. avium/intracellulare

- The differential diagnosis in those with severe combined immune deficiency is broad. For those with advanced HIV/AIDS, critically ill and ground glass, pulmonary opacities on CxR:
 - *Pneumocystitis jirovecii* pneumonia, tuberculosis, *S. pneumoniae*, *S. aureus* (including MRSA), *P. aeruginosa*, influenza, HSV, CMV, VZV, *Histoplasma*, *Coccidioides*, pulmonary lymphoma with lymphangitic spread, acute pulmonary edema, ARDS

Question

Which of the following is most common opportunistic infection in persons living with AIDS not on antiretroviral therapy?

1. Cytomegalovirus retinitis
2. *Mycobacterium avium/intracellulare*
3. Cryptococcal meningitis
4. *Pneumocystis jirovecii* pneumonia

Question

The reservoir of *Pneumocystis jirovecii* is

1. Avian droppings
2. Mammalian lungs
3. Swine intestines
4. Sneakers (running shoes)

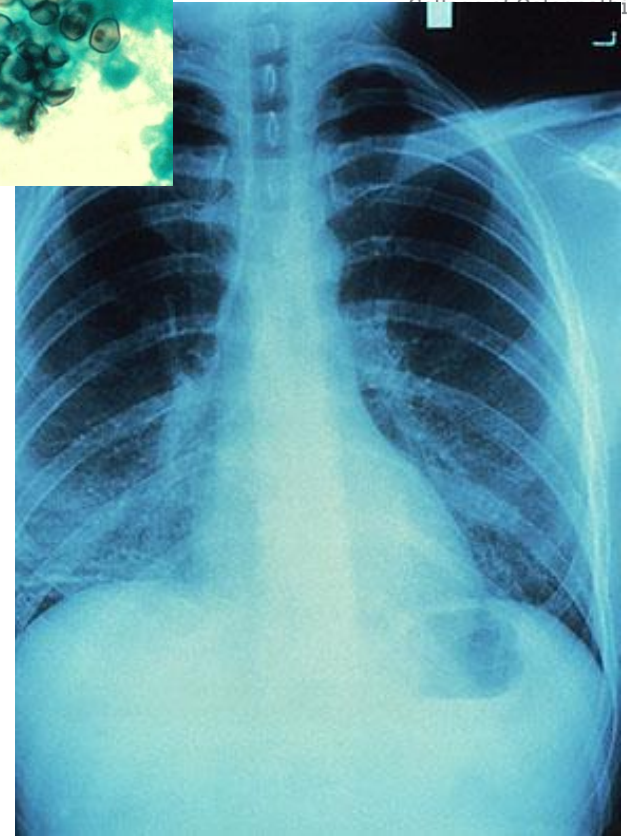
Question

Which of the following is true regarding *Pneumocystis jirovecii* pneumonia?

1. It is a “typical” pneumonia
2. Risk is increased with short course steroids
3. Pulmonary exam reveals rales
4. Sputum and hemoptysis are present
5. None of the above

Pneumocystis jirovecii

- Fungal pathogen causing widespread asymptomatic infection in mammals
- Transmission: inhalation person-to-person
- Risk: cell-mediated immune deficiency (AIDS #1, steroids)
- Presentation: CD4<200, subacute pulmonary symptoms beginning with nonproductive cough, fever followed by progressively worsening SOB
- Exam: Lungs usually quiet
- CxR: bilateral interstitial ground glass opacities without effusion
- Diagnosed by:
 - Methenamine silver stain of tissue, PCR or antigen of saliva/bronchoalveolar lavage
 - Pathology: foamy macrophages
 - Beta-D-glucan of blood, ↑LDH
- Treatment: TMP/SMZ, prednisone in severe cases. SMZ allergy: pentamidine, or if mild-moderate, atovaquone, dapsone
- Prophylaxis: TMP/SMZ, dapsone, pentamidine



Question

A 74-year-old man living with AIDS, CD4 48/ml, presents with odynophagia and weight loss? EGD shows long, deep linear esophageal ulceration. Which test would most likely result in a diagnosis?

1. Tzanck smear
2. PCR
3. Gram stain
4. Germ tube

Cytomegalovirus (CMV)

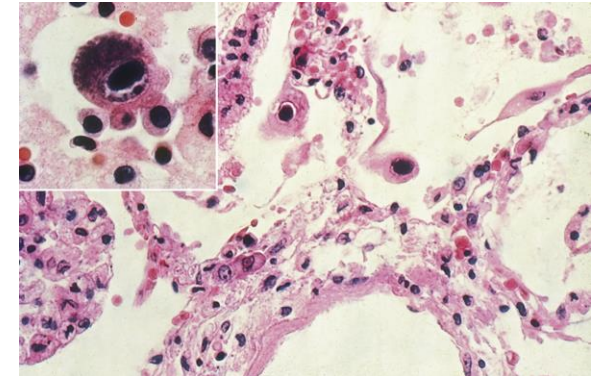
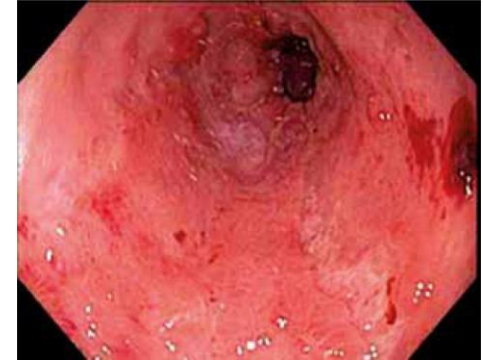
- Beta herpesviridae
 - Transmission: infection in childhood, lifelong latent infection, reactivation disease, organ donation
 - Risk: cell-mediated immune deficiency (AIDS, organ transplant), elderly
 - Presentation: CD4<50, retinitis - visual loss/floaters, flashing lights, pneumonitis –SOB, cough, ulcerative esophagitis - dysphagia, colitis – diarrhea, transplant failure
 - Exam: retinal flame hemorrhages and cotton wool spots, EGD - Long, deep, linear ulcers, CxR - bilateral ground glass opacities without effusion
- Diagnosed by:
- Clinically on retinal exam
 - Blood/tissue PCR
 - Tissue biopsy: Owl's eye inclusion bodies
- Treatment: ganciclovir (IV, intravitreal), valganciclovir (oral), cidofovir, foscarnet
 - Prophylaxis: valganciclovir

Risk of posttransplant CMV infection (1-6 months)

High risk—CMV Ab (–) recipient, CMV Ab (+) donor

Intermediate risk—CMV Ab (+) recipient, CMV Ab (+) or Ab (–) donor

Lowest risk—CMV Ab (–) recipient, CMV Ab (–) donor



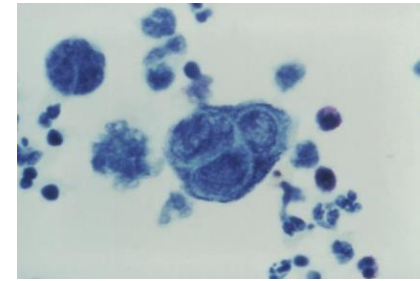
Question

A 74-year-old man living with AIDS, CD4 48/ml, presents with confusion and auditory hallucinations. MRI shows hyperintense lesion in the left temporal lobe. Transmission of the causative organism most likely occurred by which route?

1. Airborne
2. Skin-to-skin contact
3. Ingestion of spores
4. Fecal-oral

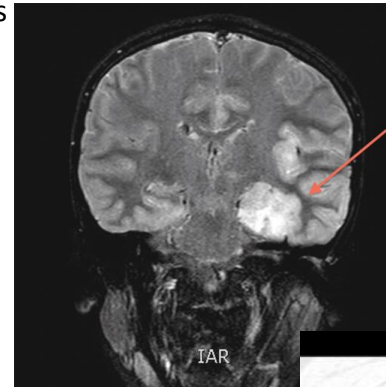
Herpes Simplex Virus (HSV)

- Alpha herpesviridae
- Transmission: infection during or after childhood, latent infection, reactivation disease
- Risk: cell-mediated immune deficiency (AIDS, organ transplant), elderly
- Presentation: CD4<50, retinitis - visual loss, acute retinal necrosis painful, chronic oral ulcers – oral pain
esophagitis – odynophagia
tracheobronchitis – chest pain,
pneumonitis – SOB, cough,
colitis - diarrhea
encephalitis – confusion, altered consciousness, hallucinations
- Exam: retinal pale peripheral lesions with central necrosis,
EGD: crops of vesicles, shallow ulcers
CT chest: bilateral ground glass opacities without effusion
- MRI brain: **hyperintensity of unilateral temporal lobe**
Diagnosed by:
 - PCR of ulcer, tissue, CSF
 - Tissue biopsy: **Multinucleate giant cells on Tzanck test**
- Treatment: acyclovir (IV, oral), foscarnet, famciclovir, valacyclovir
- Prophylaxis: acyclovir



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Source: J.E. Tintinalli, J.S. Stappczynski, G.J. Ma, D. Yealy, G.D. M.
D.M. Cline: Tintinalli's Emergency Medicine: A Comprehensive St
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(a)

Question

A 74-year-old man living with AIDS, CD4 48/ml, presents with a vesicular rash in a left T4 dermatomal distribution. There 3 similar lesions that cross the midline by ~1 cm and 5-6 more lesions are distant from the dermatome. What is the most likely causative organism?

1. Varicella zoster virus (VZV)
2. Herpes simplex virus-1 (HSV-1)
3. Herpes simplex virus-2 (HSV-2)
4. Cytomegalovirus (CMV)

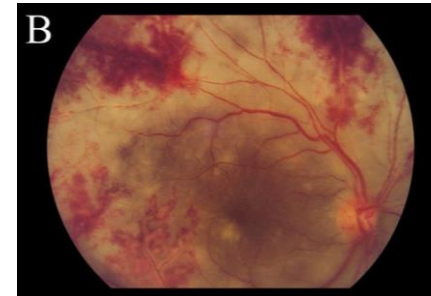
Varicella Zoster Virus (VZV)

- Alpha herpesviridae
- Transmission: infection during childhood (unvaccinated), latent infection, reactivation disease; primary infection in vaccinated, immune suppressed
- Risk: cell-mediated immune deficiency (AIDS, organ transplant), elderly
- Presentation: zoster in elderly
CD4<200, retinitis - visual loss,
CD4<50 acute retinal necrosis
pneumonitis –SOB, cough,
encephalitis – confusion, altered consciousness
- Exam: retinal diffuse flame hemorrhages
CT chest: nodular densities +/- surrounding ground glass attenuation
Diagnosed by:
 - PCR of ulcer, tissue, CSF
 - Tissue biopsy: Multinucleate giant cells on Tzanck test
- Treatment: acyclovir (IV, oral), famciclovir, valacyclovir
- Prophylaxis: acyclovir, vaccine, varicella zoster virus immune globulin



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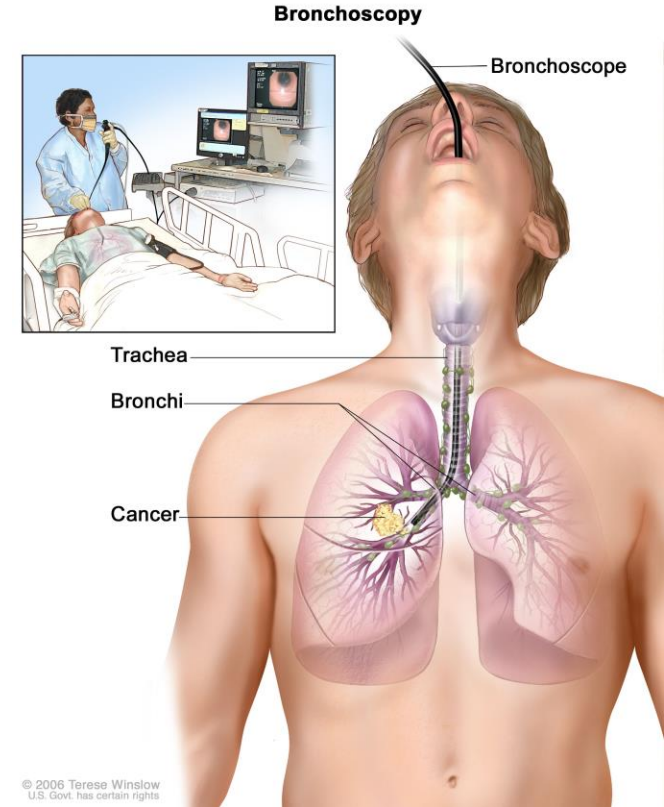
TMP/SMZ IV, prednisone, vancomycin, meropenem and fluconazole are begun.

Labs: CD4 3, HIV-1 PCR >1,000,000/ml, LDH 949 (elevated), blood cultures, beta-D-glucan and CMV PCR neg, pancytopenia, elevated transaminases, acute renal insufficiency.

Sputum by tracheal suction was *Pneumocystis* antigen positive.

Procedure?

Bronchial biopsy shows foamy macrophages. Methamine silver stain shows *Pneumocystis*-like organisms. Bacterial and fungal cultures and viral respiratory panel of bronchoalveolar lavage are negative.



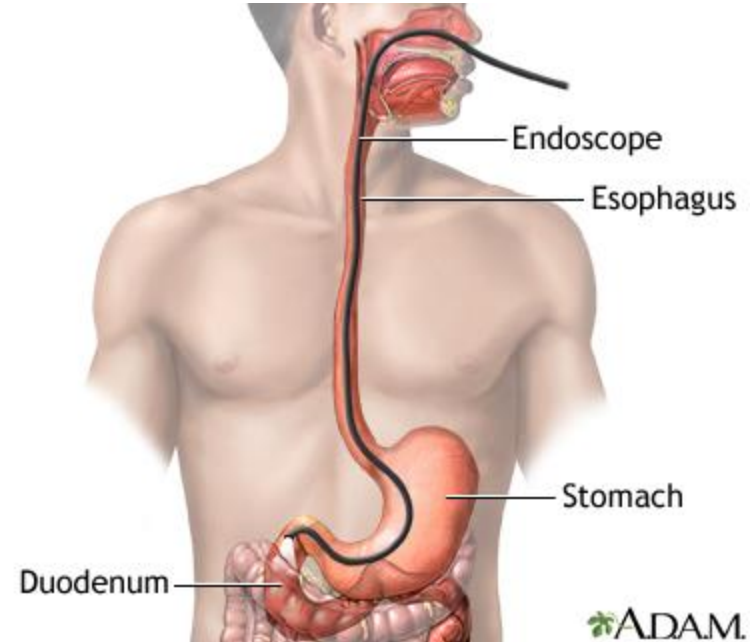
Vancomycin and meropenem are discontinued. Oxygenation improves. He is extubated, LDH normalizes and on hospital day 5 transfers to the acute ward on oxygen by nasal cannula. On hospital day 8, he complains of severe odynophagia. Oral examination reveals resolution of oral candidiasis.

What procedure is performed?

Pause

Bobby

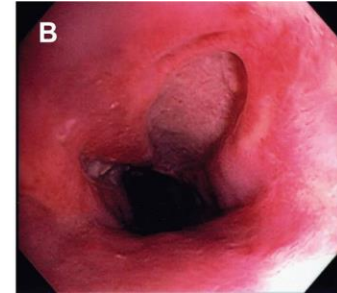
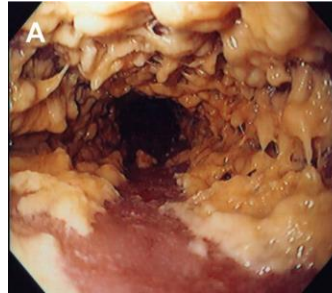
EGD* shows, deep, long, linear ulcerations in the mid esophagus.



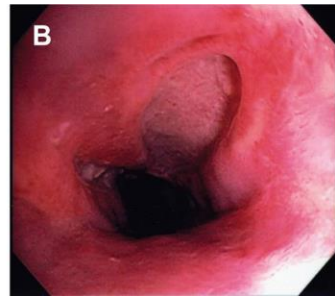
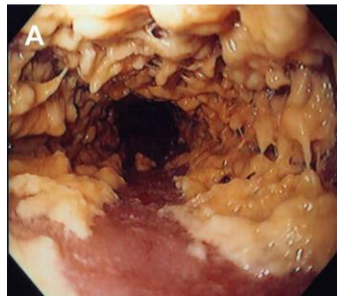
Question

Match the figures A, B and C with the causative organism.

1. Cytomegalovirus
2. Herpes simplex
3. Candida sp.



Opportunistic Esophagitis



- Odynophagia in an immune compromised host
 - A *Candida albicans*: mucoid, cottage cheese exudate
Gram stain. Germ tube. High dose fluconazole.
 - B. CMV: long deep, linear ulcer. Tzanck or PCR.
Ganciclovir IV → valganciclovir.
 - C. HSV: crops of shallow ulcers. Tzanck or PCR.
Valacyclovir, acyclovir IV/PO.

CMV PCR of an ulcer is positive. Ganciclovir IV is begun, and odynophagia improves. Three days later on hospital day 12, he has new chills associated with diaphoresis and tachycardia. He has no fever. He has also noted mild axillary submental and inguinal adenopathy. Pancytopenia persists. Alkaline phosphatase increases to 4x normal. Mycobacterial sputum and blood cultures obtained on admission are reported from the lab to be growing an organism.

What procedure is performed?

Pause

Question

A 32-year-old man living with AIDS, CD4 2/ml, presents fever, chills, adenopathy and severe anemia. Bone marrow biopsy shows organisms that stain with Ziehl-Neelsen. Treatment of this organism should start with which of the following?

1. Ethambutol and rifampin
2. Azithromycin and ethambutol
3. Clarithromycin and rifampin
4. Isoniazid and rifampin

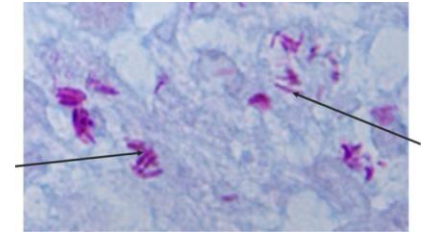
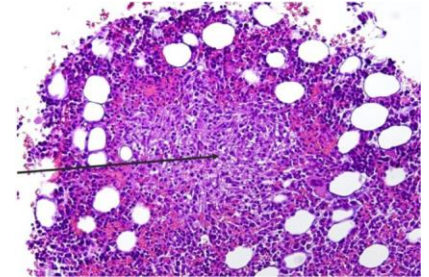
Question

Which of the following is given for primary prophylaxis for the causative organism of the last question?

1. Azithromycin
2. Clarithromycin
3. Ciprofloxacin
4. None of the above

Mycobacterium avium/intracellulare (MAI)

- Ubiquitous, water-borne organism
 - Rapidly growing, non-tuberculous mycobacteria
 - AIDS: CD4 <50
 - Fever, weight loss, diarrhea, lymphadenopathy, elevated serum alkaline phosphatase, cytopenias
 - Diagnosis mycobacterial blood culture, bone marrow biopsy, PCR, Ziehl-Neelsen stain
 - Treatment with clarithromycin or azithromycin (test for resistance!) and ethambutol
 - Prophylaxis: no longer given due to ART efficacy
 - [Lady Windermere's Syndrome – MST1R mutation](#)



Bone marrow biopsy shows organisms on Ziehl-Neelsen stain and sputum PCR is positive for MAI. Azithromycin and ethambutol are begun. TMP/SMZ, prednisone, valganciclovir are continued. On hospital day 16, he complains of a vague headache that may have been present for several days, but today is severe enough to be bothersome and he notes mild-moderate neck stiffness.

What procedure is performed?

Pause

Meningitis in AIDS

- DDx: *Cryptococcus neoformans*, *Histoplasma capsulatum*, *Coccidioides immitis*, *M. tuberculosis*, syphilis, *Listeria*, Lymphomatous, aseptic
- Fever, headache, photophobia, neck stiffness
C. neoformans may be asymptomatic
Altered mental status may indicate increased intracranial pressure, risk for uncal herniation
- Dx: Lumbar puncture

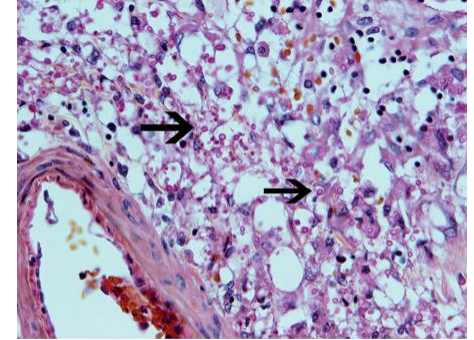
Question

The organism that most commonly causes meningitis in those living with advanced AIDS is associated with which of the following?

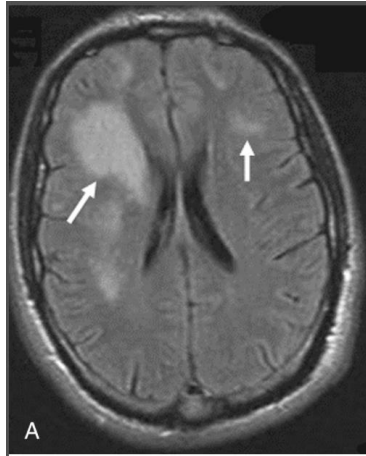
1. Travel to the southwest U.S.
2. Exposure to arboreal species
3. Acute retinal necrosis
4. Asymptomatic infection

Cryptococcus

- Yeast-like fungus derived from soil contaminated with avian droppings (*C. neoformans*) or arboreal species (*C. gattii*)
- AIDS CD4 <50/ml, hematologic malignancy, solid organ transplant, corticosteroids
 - Most common cause of meningitis in persons living with AIDS
- Inhalation of yeasts that disseminate to meninges
- Capsule is antiphagocytic and capsular polysaccharide (antigen) actively prevents immune response
- Fever, headache, photophobia, neck stiffness
- LP: ↓glucose, ↑protein, mononuclear WBCs, and opening pressure. CSF India ink, cryptococcal antigen titer
Blood: serum cryptococcal antigen
Bronchial fluids: mucicarmine or methenamine silver stains
- Treatment: liposomal amphotericin B and flucytosine followed by oral fluconazole
- Prophylaxis: fluconazole

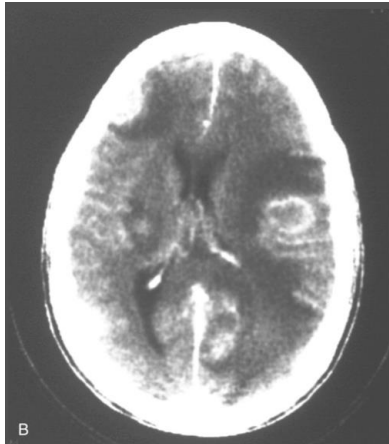


Other Cerebral Opportunistic Infections



A. Progressive Multifocal Leukoencephalopathy (PML)

- JC polyomavirus (nonenveloped, circular dsDNA)
- Immune competent: 90% are infected (asymptomatic)
Immune suppressed, (CD4 <50, natalizumab): Uni- or multifocal cerebral demyelination
- Subacute cognitive impairment, focal motor deficits, and seizures, MRI brain shows demyelinating lesions
- MRI: White matter demyelinating lesions
- DDx: AIDS dementia complex, CMV encephalitis, infarction
- Dx: CSF JC virus NAT
- Treatment: ART



B. *Toxoplasma gondii*

- Protozoan parasitic infection
- Ingestion of tissue cysts in raw or undercooked meat, mother to fetus
- Immune competent: asymptomatic, seizures from calcified cerebral cysts, chorioretinitis
Immune suppressed (CD4 <50): chorioretinitis,
- Fever, headache, confusion, motor defects, and seizures
- CT or MRI: space-occupying gray matter lesion
- DDx: lymphoma
- Dx: CSF Toxoplasma PCR, serology, pathology
- Treatment: sulfadiazine-pyrimethamine or clindamycin-pyrimethamine

Lumbar puncture shows glucose of 10 mg/dl, protein 220 mg/dl, WBC 1200/ml with 80% mononuclear cells. India ink is positive and cryptococcal antigen titer is 1:64. Opening pressure was 22 mmH₂O.

Liposomal amphotericin B and flucytosine are begun with the plan to switch to oral fluconazole in 2 weeks. His headache and meningismus improve. Discharge planning is begun and his medications at home will also include TMP/SMZ, valganciclovir, azithromycin and ethambutol.

When Bobby asks why is not going home on ART, what do you tell him?

Question

Which of the following organisms is associated with immune reconstitution inflammatory syndrome when ART is begun when the CD4 is 200?

1. *Mycobacterium avium* adenitis
2. Tuberculosis
3. Cytomegalovirus retinitis
4. *Pneumocystis jirovecii* pneumonitis

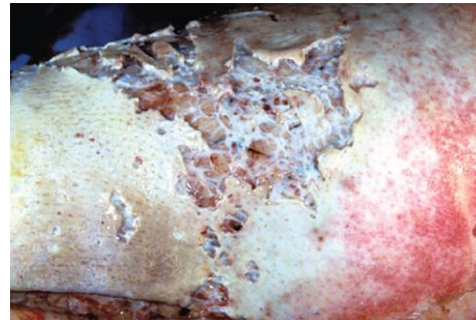
Immune Reconstitution Inflammatory Syndrome (IRIS)

- Initiation of ART results in inflammatory reaction that results in tissue injury
 - Unmasking IRIS: Undiagnosed, occult infection
 - Paradoxical IRIS: Residual antigenic stimulation
- CD4 prior to ART usually <100 (TB <200)
- Occurs in up to 13% one week to 3-4 months after ART initiation
- *Manifestations and treatment dependent on organ and organism*
 - *Mycobacterium tuberculosis*
 - *Mycobacterium avium complex*
 - Cytomegalovirus
 - *Cryptococcus neoformans*
 - *Pneumocystis jirovecii*
 - Herpes simplex virus
 - Hepatitis B virus
 - Human herpes virus 8 (associated with Kaposi sarcoma)
 - PML
- Prevention: treat known infection 2-4 weeks prior to initiating ART

Question

Match each figure with the causative infecting organism.

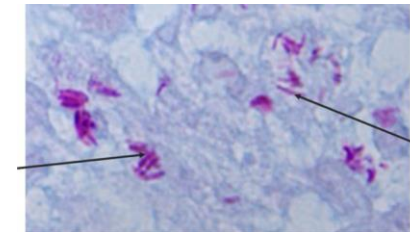
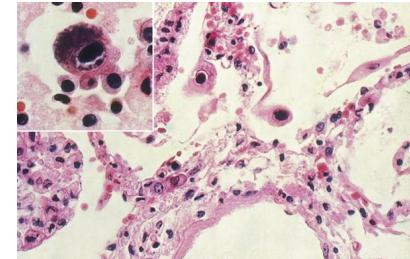
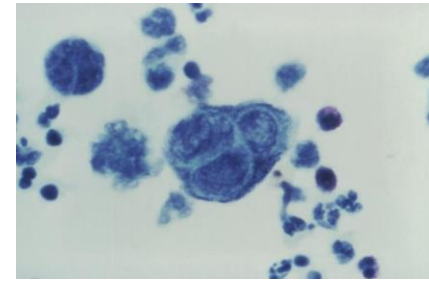
1. Human herpesvirus 8
2. *Staphylococcus epidermidis*
3. *Bartonella henselae*
4. Herpes simplex 2
5. *Pseudomonas aeruginosa*



Question

Match each figure with the causative infecting organism.

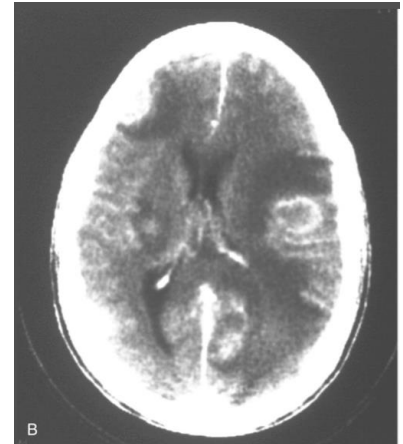
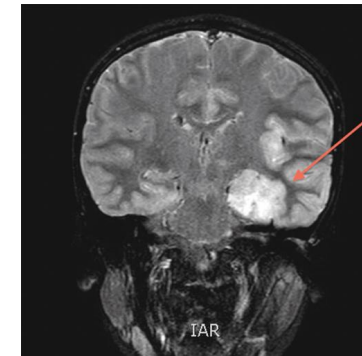
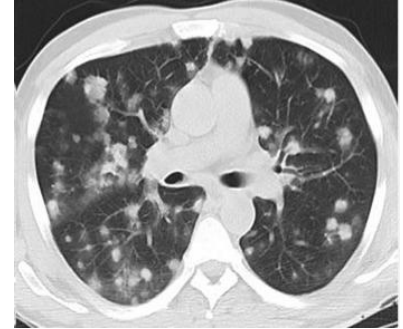
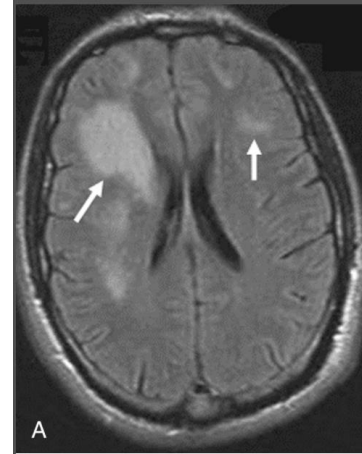
1. *Mycobacterium avium*
2. *Cryptococcus neoformans*
3. Cytomegalovirus
4. Herpes simplex 1
5. *Pneumocystis jirovecii*



Question

Match each figure with the causative infecting organism.

1. Varicella zoster virus
2. *Toxoplasma gondii*
3. JC virus
4. Herpes simplex 1
5. Lymphoma



What is the importance of an AIDS-definition?

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- Office hours: Thursday 1-2 pm CST (2-3 pm EST)
Answer keys will be released in the future for the questions in this lecture
- Lecture Feedback Form on Canvas
- Wilson Hall: Office 424C